

Notes: Agarwal and Somaini (Econometrica, 2018)
Demand Analysis Using Strategic Reports: An Application to a School Choice Mechanism

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Cambridge Controlled Choice Plan	4
3.2	Application data	4
3.3	Ranking, priority, and assignment outcomes	5
3.4	Assignment probabilities and beliefs	5
4	Methodology and empirical specifications	6
4.1	Regression-discontinuity evidence on strategic ranking	6
4.2	Preference model	6
4.3	Reports as choices over lotteries	6
4.4	Identification	7
4.5	Estimation procedure	7
4.6	Counterfactual mechanisms and welfare	7
5	Identification and interpretation	8
5.1	What the regression discontinuity identifies	8
5.2	What can be learned from strategic reports	8
5.3	Why distance is important	8
5.4	Interpreting the welfare comparison	8
5.5	Behavioral assumptions	9
5.6	Main limitations	9
6	Tables and figures	9
6.1	Tables I and II: Mechanism taxonomy and Cambridge environment	9
6.2	Tables III and IV: Raw ranking and assignment patterns	10
6.3	Figure 1: Strategic response to proximity priority	11

6.4	Table V and Figure 2: RD estimates and revealed-preference geometry	12
6.5	Figures 3 and 4: Identification intuition	13
6.6	Tables VI and VII: Assignment probabilities and estimated utilities	14
6.7	Tables VIII and IX: Truthful reporting losses and true top choices	15
6.8	Tables X and XI: Mechanism comparison and mixture model	16
6.9	Table XII: Deferred Acceptance versus Cambridge in the mixture model	17
7	Short summary	18
8	Conclusion	19
8.1	Core conclusion	19
8.2	Economic intuition	19
8.3	Incremental contribution revisited	19
8.4	Implications for school-choice policy	19
8.5	Implications for empirical work	19
8.6	Limitations	19
8.7	Takeaway	19

1 Big picture

- **Research question.** How can researchers estimate true student preferences when school-choice applications are strategic reports rather than truthful preference rankings?
- **Economic problem.** Many real-world school assignment systems are manipulable. In these systems, a rank list is not simply a statement of taste. It is also a strategic choice that trades off school desirability against admission probability.
- **Main idea.** A submitted rank-order list is treated as a choice of a lottery over school assignments. The student chooses the report whose assignment lottery gives the highest expected utility.
- **Empirical setting.** The paper studies kindergarten assignment in Cambridge, Massachusetts, from 2004–2008. Cambridge uses an Immediate Acceptance mechanism with priority for siblings and proximity to the two nearest schools.
- **Evidence of strategic behavior.** Ranking behavior changes discontinuously at proximity-priority boundaries. Students are more likely to rank a school first when they just barely have priority at that school, especially for competitive schools and paid-lunch students.
- **Methodological contribution.** The paper develops a revealed-preference approach for estimating preferences from strategic rank lists in a broad class of single-unit assignment mechanisms.
- **Main empirical finding.** Treating rank lists as truthful overstates satisfaction. About 83% of students are assigned to their stated first choice, but only about 72% are assigned to their estimated true first choice.
- **Welfare finding.** Under the baseline rational-expectations model, the average student slightly prefers the Cambridge Immediate Acceptance assignment to the Deferred Acceptance assignment by about 0.08 miles. This advantage is concentrated among paid-lunch students.

Economic takeaway. In manipulable mechanisms, stated preferences are equilibrium objects, not preference data. Demand analysis must model strategic incentives; otherwise, the researcher may confuse tactical avoidance of competitive schools with genuine tastes.

2 Incremental contribution of the paper

- **From truthful reports to strategic reports.** Much empirical work in school choice treats rank-order lists as true preferences. Agarwal and Somaini show why this can be misleading and provide a method that does not require truthful reporting.
- **A general revealed-preference framework.** The paper maps each feasible report into an assignment lottery and uses the optimality of the observed lottery to restrict the student’s utility vector.
- **Identification theory.** The paper derives conditions under which preferences can be identified from strategic reports: either rich variation in choice environments or a preference shifter such as distance to school.
- **Estimation for a broad mechanism class.** The paper gives a two-step estimator for mechanisms defined by cutoffs and priorities. The class includes Immediate Acceptance, Deferred

Acceptance with truncation, serial dictatorship with truncation, First Preferences First, and related mechanisms.

- **Large-market assignment-probability result.** The paper shows how to consistently estimate assignment probabilities even though one student’s assignment depends on all other students’ reports and priorities.
- **Behavioral flexibility.** The baseline assumes rational expectations, but the paper also estimates models with adaptive expectations, coarse beliefs, and a mixture of naive and sophisticated agents.
- **Empirical mechanism comparison.** The preference estimates are used to compare the Cambridge mechanism with student-proposing Deferred Acceptance, including assignment rates, welfare, and justified envy.
- **Policy relevance.** The paper shows that a high share of students assigned to their stated first choice does not necessarily mean that the mechanism is delivering many students to their true first choice.

3 Data and measurement

3.1 Cambridge Controlled Choice Plan

Cambridge assigns kindergarten seats using a school-choice process. Students can rank up to three programs. The mechanism is an Immediate Acceptance mechanism: first choices are considered first, then second choices, then third choices. This creates strong incentives to rank a competitive school first if the student wants to use priority there.

- **Schools and programs.** The district has 13 elementary schools. Some schools are split into multiple programs, for example bilingual or immersion programs and lunch-status-specific programs used by the assignment rule.
- **Socioeconomic quotas.** Schools are divided into free/reduced-lunch and paid-lunch programs for assignment purposes, reflecting Cambridge’s diversity goals.
- **Priority rules.** Students receive sibling priority at a sibling’s school and proximity priority at the two closest schools to their residence.
- **Tie-breaker.** A random tie-breaker ranks students within priority groups.
- **Strategic pressure.** Under Immediate Acceptance, priority is most valuable when a school is ranked first. A student can effectively lose the value of priority by ranking a competitive school second or third.

3.2 Application data

The main data cover Cambridge kindergarten assignment for school years 2004–2005 through 2008–2009.

- **Market size.** About 426 students participate in the January kindergarten lottery in an average year.

- **Capacity.** The system has about 470 seats per year.
- **Lunch status.** About 34% of applicants are free/reduced-lunch students and 66% are paid-lunch students.
- **Student characteristics.** The data include ethnicity, language spoken at home, lunch status, residential location, ranked programs, priorities, and final assignments.
- **Distance.** Distance to school is central. The average student lives about 0.49 miles from the closest school and about 1.92 miles from the average school.

3.3 Ranking, priority, and assignment outcomes

A simple reading of the raw assignment statistics would suggest high satisfaction: most students rank three programs and most students are assigned to their first-ranked program. The paper’s key point is that this interpretation is unsafe when the mechanism is strategic.

- **Maximum list use.** On average, 87% of students rank the maximum of three programs.
- **Stated first-choice assignment.** About 82% are assigned in the first round.
- **Paid-lunch versus free-lunch differences.** Paid-lunch students face a tougher market: about 76% are assigned in the first round and 15% remain unassigned, while free/reduced-lunch students have first-round assignment around 92% and almost no unassignment.
- **Priority at ranked schools.** About 51% of students have proximity priority at their first-ranked school, but only 36% and 23% have proximity priority at the second- and third-ranked schools.

Interpretation. These facts are consistent with strong preferences for nearby schools, but they are also consistent with strategic “cashing in” of proximity priority by ranking priority schools first.

3.4 Assignment probabilities and beliefs

For each report and priority type, the paper estimates the probability of assignment to each program. These probabilities are the lotteries that students choose among.

- **Rational expectations.** Students know the assignment probabilities induced by other students’ equilibrium reporting behavior.
- **Adaptive expectations.** Students use assignment probabilities based on prior-year behavior.
- **Coarse beliefs.** Students use a simplified perception of priorities and competitiveness rather than all fine distinctions in the mechanism.
- **Mixture model.** Some agents report truthfully while sophisticated agents best respond strategically.

4 Methodology and empirical specifications

4.1 Regression-discontinuity evidence on strategic ranking

The paper first shows that rank lists respond to incentives. Cambridge gives proximity priority at the two closest schools. For each student-school pair, define the boundary distance as

$$\Delta d_{ij} = d_{ij} - \frac{1}{2}(d_{i2} + d_{i3}), \quad (1)$$

where d_{i2} and d_{i3} are distances to the student's second- and third-closest schools. If $\Delta d_{ij} < 0$, the student has proximity priority at school j ; if $\Delta d_{ij} > 0$, she does not.

The empirical design compares ranking probabilities for schools just inside and just outside this priority boundary. If reports are truthful and residential location is smooth around the boundary, ranking should not jump discontinuously exactly where priority changes.

- **Baseline discontinuity.** The probability of ranking a school first falls by 5.75 percentage points at the boundary where priority is lost.
- **Competitive schools.** The first-rank effect is larger for competitive schools, at 7.27 percentage points.
- **Paid-lunch students.** Paid-lunch students show the strongest strategic response: the competitive-school first-rank discontinuity is about 11.07 percentage points.
- **Placebo test.** A placebo boundary based on the midpoint of the two closest schools shows essentially no jump.

4.2 Preference model

Student i 's utility from program j is written as

$$V(z_{ij}, \xi_j, \varepsilon_i) = U(z_{ij}^2, \xi_j, \varepsilon_i) - z_{ij}^1, \quad (2)$$

where z_{ij}^1 is the additive preference shifter. In the application, distance to school plays this role. The outside option is normalized to zero.

The specification allows for school fixed effects, student characteristics, distance costs, language-program interactions, sibling effects, and unobserved heterogeneity in school tastes.

4.3 Reports as choices over lotteries

A report R and priority type t imply an assignment lottery

$$L_{R,t} = (\Pr(1 \mid R, t), \dots, \Pr(J \mid R, t)). \quad (3)$$

A student with utility vector v_i chooses the report that maximizes expected utility:

$$R_i \in \arg \max_{R \in \mathcal{R}_i} v_i \cdot L_{R,t_i}. \quad (4)$$

This yields revealed-preference inequalities. The observed report R_i must satisfy

$$v_i \cdot (L_{R_i,t_i} - L_{R',t_i}) \geq 0 \quad \text{for all feasible } R'. \quad (5)$$

The inequalities define a normal cone in utility space. The observed report tells the econometrician that the student's utility vector belongs to that cone.

4.4 Identification

The paper studies two sources of identification.

- **Variation in choice environments.** If similar students face different lotteries because of different capacities, mechanisms, priorities, or market conditions, variation in the normal cones helps recover preference distributions.
- **Preference shifter.** If distance enters utility additively and independently of unobserved tastes, local variation in distance shifts utilities and can identify the distribution of indirect utilities.

Cambridge application. The empirical strategy uses distance as the preference shifter. This requires an assumption that, within Cambridge, unobserved preferences for schools are not driving residential sorting in a way that violates the distance-shifter condition.

4.5 Estimation procedure

The estimator has two steps.

- **Step 1: estimate assignment probabilities.** For each possible report and priority type, the paper simulates the assignment mechanism using observed market data and draws of tie-breakers. This produces estimated assignment lotteries.
- **Step 2: estimate preferences.** Given estimated lotteries, the paper estimates a random utility model. The likelihood contribution of a report is the probability that the student's utility vector lies in the cone that rationalizes the observed report.
- **Computation.** The paper uses a Gibbs sampler adapted to the linear inequality restrictions implied by optimal reports.
- **Inference.** Bootstrap standard errors account for first-step estimation error in assignment probabilities.

4.6 Counterfactual mechanisms and welfare

The paper compares the observed Cambridge mechanism to student-proposing Deferred Acceptance. Welfare is expressed in distance units: a utility difference is converted into the equivalent change in miles traveled.

- **Deferred Acceptance.** Strategy-proof for students under unrestricted lists, so students can truthfully report preferences.
- **Cambridge Immediate Acceptance.** Manipulable, but it may screen based on intensity of preferences because students strategically choose which school to rank first.
- **Justified envy.** The paper also counts cases where a student prefers another student's assignment and has higher priority at that school.

5 Identification and interpretation

5.1 What the regression discontinuity identifies

The boundary design identifies a discontinuous change in ranking behavior at the point where proximity priority changes. Since distance to school changes smoothly at the boundary, a discontinuity is evidence that students respond to priority incentives.

Interpretation. The RD result does not prove that every family is sophisticated or has correct beliefs. It shows that at least some families understand and respond to the strategic payoff of ranking priority schools first.

5.2 What can be learned from strategic reports

A strategic report does not reveal a complete ranking of schools. It reveals that one assignment lottery is preferred to other assignment lotteries. Thus, each report generates inequalities in utility space rather than an ordinal preference list.

Key insight. Even a manipulated report contains preference information because the manipulation itself reflects trade-offs between utility and admission probabilities.

5.3 Why distance is important

Distance helps identify preferences because it shifts utility in a known direction. If a student lives slightly closer to one school than another, the relative utility of those schools changes. Observed changes in report choices as distance changes can trace out the distribution of school tastes.

Limitation. This logic is vulnerable if families choose residence based on unobserved school tastes. The paper reports supporting evidence that priority boundaries do not appear to strongly affect aggregate residential patterns or house prices, but a full joint model of residential and school choice is outside the paper.

5.4 Interpreting the welfare comparison

The welfare comparison is not simply a comparison of assignment rates. Deferred Acceptance may assign fewer students to their first choice but avoid strategic complexity and justified envy. Immediate Acceptance may improve average assignment utility if strategic ranking screens intensity of preferences.

Baseline result. Under rational expectations, Cambridge slightly dominates Deferred Acceptance on average. Under truthful-report estimates, the comparison can reverse, showing why modeling strategic behavior matters.

5.5 Behavioral assumptions

The paper estimates several belief models because strategic behavior depends on what students believe about admission probabilities.

- **Rational expectations** gives the cleanest equilibrium benchmark.
- **Adaptive expectations** allows students to base beliefs on past data.
- **Coarse beliefs** allows students to simplify the priority and competitiveness environment.
- **Mixture model** allows some students to be naive truth-tellers and others to be strategic.

5.6 Main limitations

- The application relies on distance as a preference shifter and therefore abstracts from a full residential sorting model.
- The welfare comparison abstracts from the cognitive and information-acquisition costs of strategizing.
- The assignment-probability estimates rely on the observed environment being informative about students' beliefs.
- The estimates are specific to Cambridge's capacity constraints, priority rules, and distribution of preferences.

6 Tables and figures

6.1 Tables I and II: Mechanism taxonomy and Cambridge environment

Read Table I:

- The table lists school-choice mechanisms used in different districts and whether each is manipulable.
- Many observed mechanisms are manipulable, including Immediate Acceptance, truncated Deferred Acceptance, truncated serial dictatorship, First Preferences First, Chinese Parallel, Cambridge, Pan London Admissions, and some top-trading-cycle variants.
- Unrestricted Deferred Acceptance is listed as nonmanipulable and appears in Boston after 2005, Seattle during 1999–2008, and Amsterdam after 2015.

Read Table II:

- Cambridge has 13 schools and roughly 426 applicants per year in the study period.

- The average seat count is 470, but relevant capacity is split by program and lunch status.
- The sample is 34% free/reduced lunch and 66% paid lunch.
- The average student is 0.49 miles from the closest school and 1.92 miles from the average school.

Economic message. The paper studies a real assignment environment where strategic incentives are common, but the market is small enough and institutionally detailed enough to estimate assignment probabilities carefully.

TABLE I
SCHOOL CHOICE MECHANISMS^a

Mechanism	Manipulable	Examples
Immediate Acceptance	Y	Barcelona ^b , Beijing ^c , Boston (pre 2005), Charlotte-Mecklenberg ^d , Chicago (pre 2009), Denver, Miami-Dade, Minneapolis, Seattle (pre 1999 and post 2009), Tampa-St. Petersburg
Deferred Acceptance w/ Truncated Lists	Y	New York City ^e , Ghanaian Schools, various districts in England (since mid '00s)
w/ Unrestricted Lists	N	Boston (post 2005), Seattle (1999–2008), Amsterdam (post 2015) ^f
Serial Dictatorships w/ Truncated Lists	Y	Chicago (2009 onwards)
First Preferences First	Y	various districts in England (before mid '00s)
Chinese Parallel	Y	Shanghai and several other Chinese provinces ^g
Cambridge	Y	Cambridge ^h
Pan London Admissions	Y	London ⁱ
Top Trading Cycles w/ Truncated Lists	Y	New Orleans ^j
New Haven Mechanism	Y	New Haven ^k

^aSource Table 1 of Pathak and Sönmez (2008) unless otherwise stated. See several references therein for details. Other sources:

^bCalsamiglia and Güell (2017);

^cHe (2016);

^dHastings, Kane, and Staiger (2009);

^eAbdulkadiroğlu, Pathak, and Roth (2009);

^fde Haan, Gautier, Oosterbeek, and van der Klaauw (2016);

^gChen and Kesten (2013);

^h“Controlled Choice Plan” CPS, December 18, 2001;

ⁱPennell, West, and Hind (2006);

^jhttp://www.nola.com/education/index.ssf/2012/05/new_orleans_schools_say_new_pu.html accessed May 20, 2014;

^kKapor, Neilson, and Zimmerman (2017).

TABLE II
CAMBRIDGE ELEMENTARY SCHOOLS AND STUDENTS^a

Year	2004	2005	2006	2007	2008	Average
Panel A: District Characteristics						
Schools	13	13	13	13	13	13
Programs	24	25	25	27	27	25.6
Seats	473	456	476	508	438	470
Students	412	432	397	457	431	426
Free/Reduced Lunch	32%	38%	37%	29%	32%	34%
Paid Lunch	68%	62%	63%	71%	68%	66%
Panel B: Student's Ethnicity						
White	47%	47%	45%	49%	49%	47%
Black	27%	22%	24%	22%	23%	24%
Asian	17%	18%	15%	13%	18%	16%
Hispanic	9%	11%	10%	9%	9%	10%
Panel C: Language Spoken at Home						
English	72%	73%	73%	78%	81%	76%
Spanish	3%	4%	4%	4%	3%	3%
Portuguese	0%	1%	1%	1%	1%	1%
Panel D: Distances (Miles)						
Closest School	0.43	0.67	0.43	0.47	0.45	0.49
Average School	1.91	1.93	1.93	1.93	1.89	1.92

^aStudents participating in the January Kindergarten Lottery. Free/Reduced lunch based on student's application for Federal lunch subsidy.

Table II: Cambridge schools, applicants, priorities, and distances.

Table I: School-choice mechanisms and manipulability.

6.2 Tables III and IV: Raw ranking and assignment patterns

Read Table III:

- About 87% of students rank three programs, the maximum allowed.
- About 82% of students are assigned in the first round.
- Paid-lunch students face more competition: 15% are unassigned on average, compared with only 1% of free/reduced-lunch students.
- Students disproportionately rank priority schools early, especially in the first-choice slot.

Read Table IV:

- The table shows school-level ranking and rejection patterns.
- Some schools are highly competitive: Graham & Parks, Haggerty, Baldwin, Morse, Amigos, Cambridgeport, and Tobin reject some students in a typical year.

- Competition differs by lunch group. Paid-lunch students face especially high competition at several popular schools.
- A school’s competitiveness determines whether ranking it second or third is likely to be useful.

Economic message. High first-choice assignment rates do not imply truthful preference revelation. In a competitive Immediate Acceptance market, students may avoid ranking schools they like if they believe admission is unlikely.

TABLE III
RANKING, ASSIGNMENT AND PRIORITIES^a

Year	2004	2005	2006	2007	2008	Average
Panel A: Round of Assignment						
First	81%	84%	85%	83%	75%	82%
Second	8%	3%	4%	7%	5%	5%
Third	5%	2%	2%	2%	4%	3%
Unassigned	6%	11%	9%	8%	16%	10%
Panel B: Round of Assignment: Paid Lunch Students						
First	80%	77%	78%	79%	68%	76%
Second	5%	4%	5%	8%	5%	5%
Third	6%	3%	4%	2%	3%	4%
Unassigned	9%	16%	14%	11%	24%	15%
Panel C: Round of Assignment: Free Lunch Students						
First	85%	95%	98%	94%	89%	92%
Second	14%	1%	2%	4%	6%	5%
Third	2%	1%	0%	1%	4%	1%
Unassigned	0%	4%	0%	2%	1%	1%
Panel D: Number of Programs Ranked						
One	2%	6%	9%	5%	12%	7%
Two	5%	6%	9%	7%	7%	7%
Three	93%	89%	82%	88%	81%	87%
Panel E: Students With Priority at Ranked Schools						
Sibling Priority at 1st Choice	38%	34%	32%	24%	34%	32%
Sibling Priority at 2nd Choice	4%	3%	1%	2%	2%	2%
Sibling Priority at 3rd Choice	0%	2%	1%	1%	0%	1%
Proximity at 1st Choice	53%	52%	50%	51%	52%	51%
Proximity at 2nd Choice	42%	34%	37%	33%	37%	36%
Proximity at 3rd Choice	22%	24%	24%	25%	21%	23%
Panel F: Mean Distance (Miles)						
Ranked first	1.19	1.18	1.24	1.29	1.19	1.22
All ranked schools	1.37	1.41	1.38	1.40	1.34	1.38
Assigned School	1.10	1.01	1.07	1.12	0.92	1.04

^aSibling and proximity priority as reported in the Cambridge Public School assignment files. Students with older siblings enrolled in CPS receive priority at their sibling’s school. Students also receive proximity priority at their two closest schools. Percentages, where

Table III: Ranking, assignment round, and priorities.

6.3 Figure 1: Strategic response to proximity priority

Read Figure 1:

- The x-axis is distance from the boundary where proximity priority changes.
- The probability of ranking a school first drops sharply when a student loses proximity priority.
- The discontinuity is strongest for competitive schools.
- The placebo boundary does not show a comparable discontinuity.

Economic message. Students act as if priority has strategic value. This is direct evidence against treating all rank lists as truthful preference orderings.

TABLE IV
RANKING AND ASSIGNMENT BY SCHOOL^a

	Graham & Parks	Haggerty	Baldwin	Morse	Amigos	Cambridgeport	King Open	Peabody	Tobin	Fletcher Maynard	Kenn Long	MLK	King Open Ols
Panel A: All Students													
Ranked First	60	56	53	47	37	34	33	31	25	18	16	12	5
Ranked Second	72	37	66	25	18	44	39	38	17	10	18	20	0
Ranked Third	56	33	46	31	19	44	37	32	20	15	16	15	0
Ranked Anywhere	192	120	166	102	75	113	114	105	64	48	54	51	6
Capacity	41	41	41	42	41	27	51	48	35	38	41	37	15
First Rejected	1-P	1-R	1-R	1-R	1-R	1-R	NR	NR	1-R	NR	NR	NR	NR
Panel B: Paid Lunch Students													
Ranked First	49	45	40	29	25	24	25	17	13	4	7	4	2
Ranked Second	60	28	56	14	12	29	23	27	10	3	6	6	0
Ranked Third	47	29	33	19	15	34	24	18	11	4	8	10	0
Ranked Anywhere	152	95	128	60	51	87	70	65	33	9	21	20	3
Capacity	29	27	27	29	41	18	36	34	29	35	34	27	15
First Rejected	1-P	1-R	1-R	1-R	1-R	1-R	NR	NR	3-R	NR	NR	NR	NR
Panel C: Free Lunch Students													
Ranked First	9	12	12	17	12	11	13	10	12	16	10	9	2
Ranked Second	13	8	7	11	5	12	17	12	8	8	14	11	0
Ranked Third	10	4	9	10	4	12	13	13	9	10	11	4	0
Ranked Anywhere	29	24	25	40	20	36	44	38	31	36	34	25	2
Capacity	25	23	26	26	41	17	33	31	19	18	26	24	15
First Rejected	NR	NR	NR	1-R	1-R	2-P	NR	NR	1-R	NR	NR	NR	NR

^aMedian number of applicants and seats over the years 2003–2008. First rejected is the round and priority of the first rejected student, e.g., 1-P indicates that a student with proximity priority was rejected in the first round. S: Sibling priority, PS: both sibling and sibling priority, R: regular priority, and NR: no student was rejected in any round. Free/Reduced lunch based on student’s application for Federal lunch subsidy.

Table IV: Ranking and assignment by school.

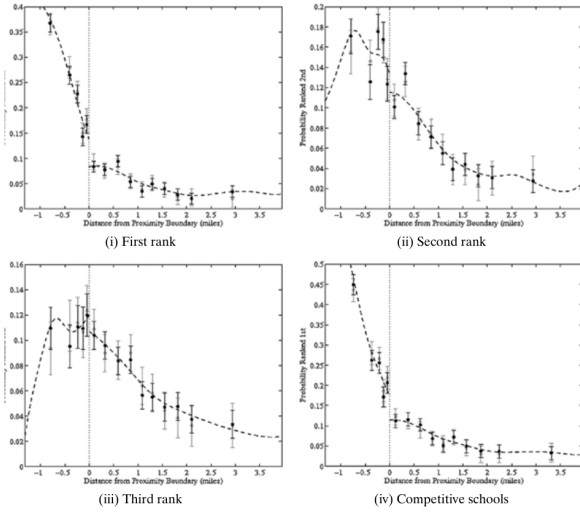


FIGURE 1. Effect of proximity-priority on ranking behavior. Notes: The sample includes students who passed

Figure 1, panels i–iv: ranking response at the proximity-priority boundary.

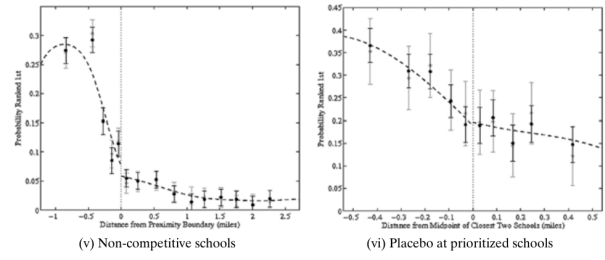


FIGURE 1.—Continued.

Figure 1, panels v–vi: noncompetitive schools and placebo boundary.

6.4 Table V and Figure 2: RD estimates and revealed-preference geometry

Read Table V:

- The baseline first-rank discontinuity is -5.75% for all students.
- For competitive schools, the discontinuity is -7.27% overall and -11.07% for paid-lunch students.
- The second- and third-rank discontinuities are smaller.
- The placebo boundary estimate is essentially zero.

Read Figure 2:

- Each report corresponds to a lottery over assignments.
- A student chooses the lottery with the highest expected utility.
- The optimality of a chosen lottery defines a cone of utility vectors that rationalize that report.
- This is the basic revealed-preference object used for estimation.

Economic message. The RD estimates motivate the strategic model, and Figure 2 shows why strategic reports are still informative: they reveal which assignment lottery a student prefers.

TABLE V
REGRESSION DISCONTINUITY ESTIMATES^a

	Baseline			Competitive School	Non-Competitive School	Placebo Boundary
	Rank First	Rank Second	Rank Third	Rank First	Rank First	Rank First
Panel A: All Students						
Estimate	-5.75%	-2.38%	-0.86%	-7.27%	-2.06%	0.07%
	(0.013)	(0.012)	(0.011)	(0.018)	(0.019)	(0.024)
t-statistic	-4.54	-2.02	-0.80	-3.96	-1.10	0.03
Panel B: Paid Lunch Students						
Estimate	-7.44%	-2.65%	-0.68%	-11.07%	-1.22%	1.88%
	(0.016)	(0.014)	(0.015)	(0.025)	(0.018)	(0.031)
t-statistic	-4.64	-1.90	-0.46	-4.45	-0.67	0.61
Panel C: Free Lunch Students						
Estimate	-3.55%	-2.59%	-3.15%	-1.47%	-5.23%	-3.55%
	(0.022)	(0.021)	(0.022)	(0.031)	(0.031)	(0.033)
t-statistic	-1.60	-1.22	-1.43	-0.47	-1.67	-1.06

^aRegression discontinuity estimates based bandwidth selection rule proposed by Imbens and Kalyanaraman (2011). All estimates use rankings by 2,128 students. Competitive schools are Graham & Parks, Haggerty, Baldwin, Morse, Amigos, Cambridgeport, and Tobin. Placebo boundary at the mid-point of the two-closest schools. Standard errors clustered at the student level in parentheses.

Table V: Regression-discontinuity estimates.

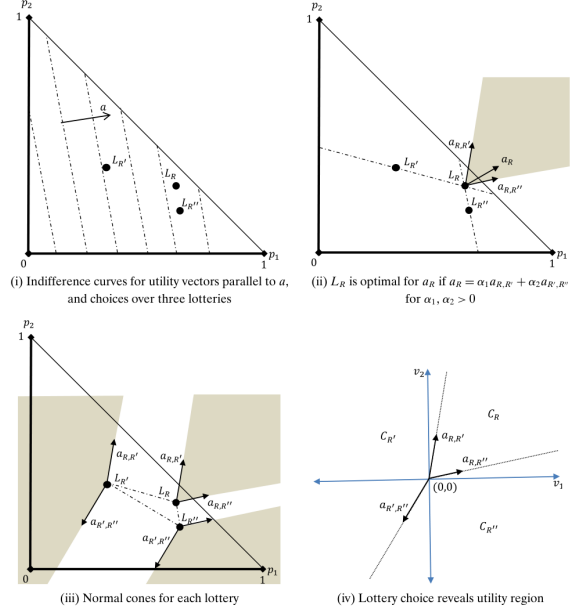


FIGURE 2.—A revealed preference argument.

Figure 2: Revealed-preference geometry for lottery choice.

6.5 Figures 3 and 4: Identification intuition

Read Figure 3:

- Variation in assignment probabilities shifts the normal cones associated with reports.
- This variation can identify the density of utility types over regions of utility space.

Read Figure 4:

- Local variation in a preference shifter z identifies the density of indirect utility u .
- In the application, distance is the main preference shifter.

Economic message. Identification does not require observing truthful rankings. It comes from how report choices vary as lotteries and distance-shifted utilities vary.

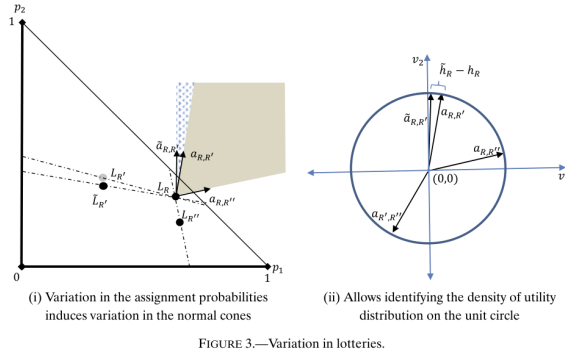


Figure 3: Variation in lotteries and normal cones.

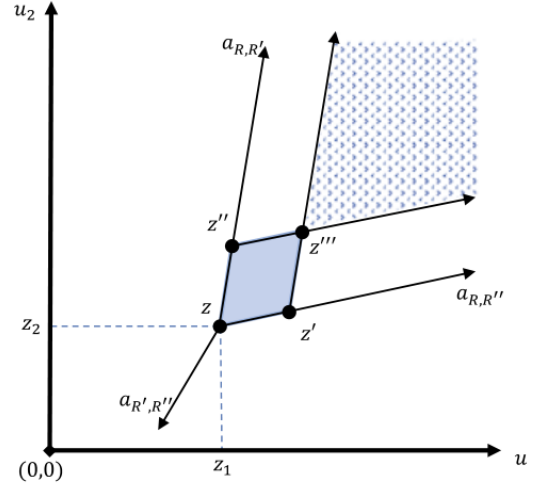


FIGURE 4.—Local variation in z identifies the density of

Figure 4: Local variation in distance-like shifters.

6.6 Tables VI and VII: Assignment probabilities and estimated utilities

Read Table VI:

- Assignment probabilities vary sharply across schools, ranks, lunch groups, and priority status.
- Paid-lunch students have only a 0.22 probability of assignment to Graham & Parks when ranking it first, versus 0.82 for free/reduced-lunch students.
- Some schools are nearly sure admission when ranked first, while others remain risky even in the first slot.
- Proximity priority substantially raises assignment probabilities, especially for first-ranked schools.

Read Table VII:

- The table reports mean school utilities net of distance under truthful, rational-expectations, adaptive-expectations, and coarse-beliefs models.
- Strategic models imply that some competitive schools are more attractive than raw ranks suggest, because students strategically avoid them.
- Graham & Parks, Haggerty, and Baldwin are valued highly by paid-lunch students in the strategic specifications.
- The estimated fraction of acceptable schools differs markedly across belief assumptions and lunch groups.

Economic message. Assignment probabilities are not background details; they are part of the choice set. Correcting for them changes estimated school demand.

TABLE VI
ESTIMATED ASSIGNMENT PROBABILITIES*

Ranked	Graham & Parks	Haggerty	Baldwin	Morse	Amigos	Cambridgeport	King Open	Peabody	Tobin	Tobin K4	Tobin K5	Fletcher	Maynard	Kenn Long	MLK	King Open Ola
First	0.43	0.59	0.63	0.57	0.73	0.98	0.60	1.00	0.94	0.85	0.31	0.34	0.02	1.00	1.00	1.00
Second	0.24	0.25	0.23	0.20	0.35	0.94	0.18	0.92	0.83	0.74	0.04	0.14	0.96	1.00	0.99	1.00
Third	0.21	0.19	0.18	0.10	0.25	0.83	0.10	0.67	0.61	0.66	0.02	0.08	0.77	0.90	0.90	0.89
First	0.22	0.45	0.49	0.54	0.73	1.00	0.51	1.00	0.94	0.93	0.32	0.36	1.00	1.00	1.00	1.00
Second	0.00	0.05	0.03	0.16	0.35	1.00	0.08	0.89	0.82	0.76	0.03	0.16	1.00	1.00	1.00	1.00
Third	0.00	0.01	0.00	0.06	0.24	0.85	0.01	0.56	0.56	0.64	0.01	0.09	0.89	0.89	0.89	0.87
First	0.82	0.87	0.90	0.64	0.74	0.97	0.77	1.00	0.94	0.72	0.31	0.29	0.76	1.00	1.00	1.00
Second	0.71	0.65	0.61	0.26	0.25	0.90	0.39	0.98	0.86	0.71	0.07	0.08	0.59	0.99	0.99	1.00
Third	0.62	0.56	0.52	0.18	0.27	0.83	0.28	0.87	0.70	0.68	0.03	0.05	0.53	0.92	0.92	0.94
First	0.63	0.97	0.95	0.89	0.95	0.99	0.94	1.00	1.00	0.92	0.55	0.54	0.95	1.00	1.00	1.00
Second	0.12	0.21	0.15	0.24	0.38	0.97	0.30	0.97	0.84	0.76	0.07	0.16	0.80	1.00	1.00	1.00
Third	0.10	0.11	0.06	0.12	0.28	0.84	0.16	0.76	0.66	0.67	0.02	0.12	0.72	0.89	0.90	0.88
First	0.37	0.55	0.60	0.52	0.71	0.97	0.55	1.00	0.94	0.85	0.28	0.33	0.92	1.00	1.00	1.00
Second	0.28	0.25	0.23	0.19	0.34	0.96	0.17	0.92	0.83	0.75	0.03	0.14	0.88	1.00	1.00	1.00
Third	0.25	0.20	0.18	0.10	0.24	0.85	0.10	0.66	0.60	0.66	0.02	0.08	0.78	0.90	0.90	0.89

*Average estimates weighted by number of students of each type. Probabilities estimated using $R = 1000$. Ranks and priority types of opposing students are drawn with replacement from the

Table VI: Estimated assignment probabilities.

TABLE VII
ESTIMATED MEAN UTILITIES*

	Truthful		Rational Expectations		Adaptive Expectations		Coarse Beliefs	
	Paid Lunch	Free Lunch	Paid Lunch	Free Lunch	Paid Lunch	Free Lunch	Paid Lunch	Free Lunch
Panel A: Mean utility								
Graham & Parks	1.29 (0.06)	0.40 (0.08)	1.93 (0.27)	-0.20 (0.33)	1.89 (0.36)	-0.19 (0.52)	1.85 (0.12)	0.47 (0.16)
Haggerty	1.39 (0.07)	0.72 (0.11)	1.45 (0.16)	0.69 (0.24)	1.41 (0.20)	0.89 (0.31)	1.56 (0.13)	0.68 (0.18)
Baldwin	1.26 (0.05)	0.50 (0.09)	1.36 (0.14)	0.70 (0.16)	1.05 (0.17)	-0.14 (0.37)	1.57 (0.10)	0.72 (0.15)
Morse	0.66 (0.07)	0.70 (0.08)	0.77 (0.12)	0.97 (0.17)	0.61 (0.18)	0.78 (0.33)	0.82 (0.14)	0.8 (0.16)
Amigos	-0.01 (0.13)	-0.38 (0.15)	0.13 (0.18)	-0.29 (0.27)	0.05 (0.22)	-0.43 (0.32)	-0.11 (0.19)	-0.64 (0.27)
Cambridgeport	0.77 (0.06)	0.18 (0.08)	0.60 (0.14)	0.29 (0.21)	0.41 (0.26)	0.25 (0.12)	0.91 (0.12)	0.21 (0.16)
King Open	0.65 (0.06)	0.40 (0.07)	0.58 (0.10)	0.27 (0.13)	0.44 (0.13)	0.37 (0.18)	0.52 (0.10)	0.24 (0.12)
Peabody	0.22 (0.08)	0.48 (0.09)	0.06 (0.12)	0.30 (0.15)	0.03 (0.16)	0.52 (0.20)	0.05 (0.11)	0.31 (0.15)
Tobin	-0.49 (0.11)	0.64 (0.12)	-0.92 (0.21)	0.38 (0.26)	-0.83 (0.25)	0.39 (0.30)	-0.74 (0.18)	0.28 (0.21)
Fletcher Maynard	-1.30 (0.14)	-0.05 (0.10)	-2.03 (0.42)	-0.14 (0.23)	-1.67 (0.32)	0.12 (0.22)	-2.21 (0.30)	-0.3 (0.18)
Kenn Long	-0.19 (0.09)	0.47 (0.07)	-0.40 (0.20)	0.11 (0.17)	-0.54 (0.27)	0.08 (0.21)	-0.51 (0.18)	0.25 (0.15)
MLK	-0.66 (0.10)	0.08 (0.09)	-1.14 (0.23)	-0.28 (0.19)	-0.82 (0.22)	0.16 (0.22)	-1.24 (0.18)	-0.27 (0.17)
King Open Ola	-3.60 (0.35)	-4.13 (0.39)	-2.39 (0.53)	-2.79 (0.67)	-2.01 (0.62)	-2.79 (0.93)	-2.47 (0.47)	-2.75 (0.63)
Outside Option	-2.08 (0.10)	-1.44 (0.09)	-0.55 (0.08)	-0.92 (0.08)	-0.64 (0.09)	-0.74 (0.13)	-0.49 (0.06)	-0.73 (0.07)
Panel B: Percentage of acceptable schools								
Up to 1	10%	20%	24%	14%	23%	19%	16%	21%
Up to 2	16%	30%	61%	40%	61%	45%	58%	51%
Up to 3	23%	40%	85%	62%	86%	69%	84%	74%
Up to 4	34%	51%	95%	79%	95%	85%	95%	89%
Up to 5	44%	61%	98%	91%	99%	95%	99%	96%

*Average estimated utility for each school, normalizing the mean utility of the inside options to zero. Utilities calculated by averaging the predicted utility given the non-distance covariates. Bootstrap standard errors in parentheses, except for Truthful reporting where we present the standard deviation of the posterior. Adaptive Expectations based on reported lists from 2005, 2006, and 2008

Table VII: Estimated mean utilities under alternative belief models.

6.7 Tables VIII and IX: Truthful reporting losses and true top choices

Read Table VIII:

- Under the truthful-report specification, 57% of students incur no loss from truthful reporting, while the average loss is 0.18 miles.
- Paid-lunch students incur larger truthful-report losses than free-lunch students: 0.26 miles versus 0.01 miles in the truthful specification.
- Under the rational-expectations estimates, about 46% have no loss from truthful reporting and the average loss is 0.07 miles.

Read Table IX:

- About 83.4% of students are assigned to the school they ranked first.
- Only about 72.3% are assigned to their estimated true top-choice school.
- The gap is especially large for paid-lunch students: 79.4% are assigned to their ranked first choice, but only 64.6% to their estimated true first choice.
- Free-lunch students face a smaller gap because they face less competition.

Economic message. Stated first-choice success exaggerates true first-choice success. Strategic avoidance makes the mechanism look more satisfying than it is if the analyst reads reports literally.

TABLE VIII
LOSSES FROM TRUTHFUL REPORTS^a

	Truthful						Rational Expectations					
	No Loss		Mean Loss		Std Loss		No Loss		Mean Loss		Std Loss	
	Mean	s.e.	Mean	s.e.	Mean	s.e.	Mean	s.e.	Mean	s.e.	Mean	s.e.
All	57%	0.01	0.18	0.02	0.53	0.05	46%	0.01	0.07	0.01	0.26	0.03
Free Lunch	68%	0.02	0.01	0.00	0.09	0.03	62%	0.02	0.01	0.00	0.07	0.03
Paid Lunch	51%	0.01	0.26	0.03	0.64	0.06	38%	0.01	0.10	0.01	0.31	0.04
Black	65%	0.02	0.06	0.02	0.30	0.07	56%	0.02	0.04	0.01	0.19	0.05
Asian	56%	0.03	0.19	0.04	0.56	0.09	46%	0.03	0.07	0.02	0.25	0.06
Hispanic	60%	0.04	0.10	0.03	0.36	0.09	51%	0.03	0.04	0.01	0.18	0.06
White	52%	0.01	0.24	0.03	0.62	0.06	40%	0.02	0.09	0.01	0.30	0.04
Other Race	47%	0.06	0.20	0.07	0.51	0.15	39%	0.05	0.08	0.04	0.24	0.09

^aEstimated loss from reporting preferences truthfully, relative to optimal report in distance units (miles).

Table VIII: Estimated losses from truthful reporting.

TABLE IX
RANKING AND ASSIGNMENT AT TOP CHOICE^a

	Graham & Parks	Haggerty	Baldwin	Moss	Amigo	Cambridgeport	King Open	Peabody	Tobin	Fletcher	Maynard	Kenn Long	MLK	King Open Ols	Total
Panel A: All Students															
*referred School	24.8	11.6	9.4	10.0	7.9	6.1	7.2	6.1	5.4	4.0	3.6	2.2	1.0	99.3	
*ranked #1 (simul)	16.2	12.6	11.6	11.2	8.4	8.0	8.6	7.0	4.9	3.9	3.6	2.3	1.0	99.3	
*ranked #1 (data)	14.3	12.6	11.9	11.0	8.8	7.7	8.2	7.8	5.7	4.4	3.8	2.7	1.2	100.0	
*referred and Ranked #1	15.3	10.4	8.2	9.6	7.5	5.8	7.2	6.1	4.6	3.9	3.6	2.2	1.0	85.2	
*referred and Assigned	9.9	8.5	6.6	8.2	6.8	5.1	7.2	6.0	3.8	3.5	3.6	2.2	1.0	72.3	
*ranked #1 and Assigned	10.4	10.1	9.0	9.5	7.6	6.8	8.6	6.9	4.2	3.6	3.6	2.3	1.0	83.4	
Panel B: Free Lunch Students															
*referred School	9.0	8.1	6.5	12.5	6.9	6.8	7.2	7.9	10.7	10.8	7.1	4.6	1.6	99.5	
*ranked #1 (simul)	8.9	8.5	6.8	12.7	6.9	7.1	7.2	8.2	9.0	10.6	7.2	4.7	1.6	99.5	
*ranked #1 (data)	6.7	8.3	8.0	12.2	7.8	6.4	7.7	9.0	8.5	10.8	7.1	5.5	2.0	100.0	
*referred and Ranked #1	8.6	8.0	6.4	12.2	6.7	6.6	7.2	7.8	8.9	10.3	7.1	4.6	1.6	96.1	
*referred and Assigned	7.9	7.5	6.0	10.7	6.0	6.1	7.2	7.8	7.1	9.3	7.1	4.6	1.6	88.8	
*ranked #1 and Assigned	8.3	7.9	6.4	11.1	6.2	6.5	7.2	8.1	7.2	9.5	7.2	4.7	1.6	91.9	
Panel C: Paid Lunch Students															
*referred School	32.6	13.3	10.9	8.7	8.6	5.7	7.1	5.3	2.7	0.6	1.8	1.1	0.7	99.3	
*ranked #1 (simul)	20.0	14.6	13.9	10.4	9.3	8.3	9.3	6.5	2.9	0.6	1.8	1.1	0.7	99.3	
*ranked #1 (data)	18.8	14.6	13.4	10.6	9.5	7.9	8.7	7.2	3.7	1.2	2.2	1.3	0.9	100.0	
*referred and Ranked #1	18.9	11.5	9.2	8.2	8.1	5.3	7.3	5.3	2.4	0.6	1.8	1.1	0.7	80.2	
*referred and Assigned	11.0	8.9	7.0	6.9	7.3	4.6	7.3	5.2	2.2	0.6	1.8	1.1	0.7	64.6	

Table IX: Stated first choice versus estimated true top choice.

6.8 Tables X and XI: Mechanism comparison and mixture model

Read Table X:

- Under rational expectations, Cambridge assigns 72.3% of students to their true first choice, compared with 67.8% under Deferred Acceptance.
- The mean utility difference $DA - Cambridge$ is -0.078 miles for all students under rational expectations, meaning Cambridge is preferred on average.
- Paid-lunch students drive the gain: the rational-expectations mean difference is -0.107 miles for paid-lunch students and only -0.021 for free-lunch students.
- Under the truthful-report model, the sign reverses, illustrating why strategic modeling matters.

Read Table XI:

- The mixture model estimates mean utilities under a model with naive and sophisticated agents.
- About a third of paid-lunch and free-lunch students are estimated to be naive truth-tellers.
- Mean utilities are broadly similar to the earlier strategic specifications.

Economic message. Cambridge can outperform Deferred Acceptance in assignment utility because strategic ranking screens intensity of preferences. But this result depends on the model of beliefs and behavior.

DEFERRED ACCEPTANCE VERSUS CAMBRIDGE ^a																
	Truthful				Rational Expectations				Course Beliefs				Adaptive Expectations			
	All Students	Paid Lunch	Free Lunch	All Students	Paid Lunch	Free Lunch	All Students	Paid Lunch	Free Lunch	All Students	Paid Lunch	Free Lunch	All Students	Paid Lunch	Free Lunch	All Students
Panel A: Deferred Acceptance																
Percent Assigned to First Choice	71.0	64.0	84.8	67.8	58.4	86.4	70.4	62.0	86.9	69.0	58.0	88.4				
Percent Assigned to Second Choice	12.0	13.6	8.7	15.8	18.7	10.0	12.3	14.1	8.8	13.8	17.3	7.5				
Percent Assigned to Third Choice	5.1	7.0	1.4	5.2	7.1	1.5	4.9	6.8	1.1	5.1	7.2	1.3				
Percent Assigned to Fourth Choice	3.0	4.4	0.3	1.3	1.9	0.2	1.8	2.6	0.1	1.5	2.3	0.2				
Percent Assigned to Fifth Choice	1.7	2.5	0.1	0.2	0.3	0.0	0.4	0.6	0.0	0.2	0.4	0.0				
Panel B: Cambridge Mechanism																
Percent Assigned to First Choice	78.9	74.5	87.7	72.3	63.9	88.8	73.9	67.3	86.9	72.2	63.0	88.9				
Percent Assigned to Second Choice	6.5	6.8	5.9	14.7	18.1	7.9	10.2	11.1	8.3	11.9	15.1	6.1				
Percent Assigned to Third Choice	3.1	4.0	1.3	3.9	5.1	1.3	3.5	4.6	1.5	3.6	4.7	1.5				
Percent Assigned to Fourth Choice	0.0	0.0	0.0	1.0	1.4	0.3	1.5	2.1	0.3	1.2	1.7	0.3				
Percent Assigned to Fifth Choice	0.0	0.0	0.0	0.2	0.2	0.0	0.4	0.5	0.0	0.2	0.3	0.1				
Panel C: Deferred Acceptance vs. Cambridge																
Mean Utility DA – Cambridge	0.005	0.102	-0.008	-0.077	-0.107	-0.021	-0.035	-0.052	-0.001	-0.032	-0.069	0.033				
	(0.019)	(0.027)	(0.008)	(0.009)	(0.012)	(0.009)	(0.009)	(0.012)	(0.011)	(0.025)	(0.033)	(0.032)				
Std. Utility DA – Cambridge	0.149	0.168	0.044	0.109	0.120	0.066	0.095	0.104	0.061	0.178	0.124	0.233				
Percent DA > Cambridge	26.1	28.0	22.2	17.3	15.6	20.6	25.2	24.6	26.3	25.0	23.4	27.8				
	(1.6)	(1.9)	(2.4)	(1.8)	(2.2)	(2.4)	(1.5)	(1.8)	(2.2)	(2.7)	(3.5)	(3.4)				
Percent DA < Cambridge	32.4	27.4	42.4	31.2	28.0	37.5	32.1	28.2	30.7	31.6	26.3	41.0				
	(1.2)	(1.5)	(2.0)	(1.2)	(1.5)	(2.0)	(1.2)	(1.4)	(1.9)	(1.7)	(1.9)	(2.8)				
Percent DA = Cambridge	41.5	44.6	35.4	51.5	56.4	41.9	42.8	47.2	34.0	43.5	50.3	31.2				
	(1.8)	(2.1)	(2.5)	(2.0)	(2.4)	(2.7)	(1.7)	(2.0)	(2.4)	(2.8)	(3.7)	(3.5)				
Percent With Justified Envy	7.7	10.4	2.4	2.5	2.7	2.3	5.0	5.8	3.4	3.8	4.2	2.9				
	(0.5)	(0.7)	(0.4)	(0.2)	(0.3)	(0.4)	(0.3)	(0.4)	(0.4)	(0.7)	(1.0)	(0.6)				

^a Notes: Panels A and B present percentages of students assigned to their true first choice. Panel C compares the expected utility difference between Deferred Acceptance and Cambridge. All numbers are rounded to the nearest tenth of a percentage point.

^aNotes: Panel A and B present percentages of students assigned to their true first choice. Panel C compares the expected utility difference between Deferred Acceptance and Cambridge Mechanism. Estimates of the Predicted Utility Difference are shown in parentheses. We report Cambridge estimates only for comparison.

Table X: Deferred Acceptance versus Cambridge.

TABLE XI
ESTIMATED MEAN UTILITIES USING A MIXTURE MODEL^a

	Mixture Model		
	All Students	Paid Lunch	Free Lunch
Panel A: Mean Utility			
Graham & Parks	1.19 (0.11)	1.53 (0.12)	0.52 (0.15)
Haggerty	1.27 (0.14)	1.53 (0.13)	0.76 (0.22)
Baldwin	1.25 (0.10)	1.45 (0.10)	0.84 (0.13)
Morse	0.74 (0.11)	0.68 (0.11)	0.86 (0.13)
Amigos	-0.12 (0.21)	0.00 (0.19)	-0.38 (0.30)
Cambridgeport	0.56 (0.11)	0.68 (0.11)	0.31 (0.15)
King Open	0.48 (0.09)	0.57 (0.10)	0.32 (0.12)
Peabody	0.08 (0.13)	0.03 (0.13)	0.16 (0.16)
Tobin	-0.45 (0.16)	-0.81 (0.18)	0.26 (0.23)
Fletcher Maynard	-1.02 (0.24)	-1.52 (0.30)	-0.04 (0.18)
Kenn Long	-0.07 (0.14)	-0.23 (0.16)	0.24 (0.15)
MLK	-0.68 (0.14)	-0.97 (0.17)	-0.10 (0.15)
King Open Ola	-3.23 (0.40)	-2.96 (0.43)	-3.77 (0.46)
Outside Option	-1.11 (0.09)	-1.03 (0.09)	-1.26 (0.11)
Panel B: Agent Behavior			
Fraction Naïve		0.378 [0.0079]	0.316 [0.0079]

^aPanel A presents the average estimated utility for each school, normalizing the mean utility of the inside options to zero. Utilities calculated by averaging the predicted utility given the non-distance covariates. Panel B reports the estimated fraction of naïve agents by free-lunch status. Bootstrap standard errors in parentheses.

Table XI: Estimated utilities in the mixture model.

6.9 Table XII: Deferred Acceptance versus Cambridge in the mixture model

Read Table XII:

- The mixture model separates naïve and sophisticated agents.
- Naïve students rank their true favorite first, which can help them at competitive top-choice schools because they do not strategically avoid them.
- Under the mixture model, the average naïve student still prefers Cambridge to Deferred Acceptance by about 0.024 miles.
- Sophisticated students prefer Cambridge by more, about 0.094 miles.
- Paid-lunch sophisticated students benefit the most in expected assignment utility.

Economic message. Naivete is not always privately disastrous in this setting because naïve students may gain priority at their true first choice. But this comes with distributional consequences and more justified envy for some groups.

TABLE XII
DEFERRED ACCEPTANCE VERSUS CAMBRIDGE USING A MIXTURE MODEL^a

	All Students		Paid Lunch		Free Lunch	
Panel A: Deferred Acceptance						
Percent Assigned to First Choice	70.0		61.8		86.3	
Percent Assigned to Second Choice	13.1		14.5		10.3	
Percent Assigned to Third Choice	5.8		7.8		1.7	
Percent Assigned to Fourth Choice	2.8		4.0		0.3	
Percent Assigned to Fifth Choice	0.9		1.4		0.1	
	Naïve	Sophisticated	Naïve	Sophisticated	Naïve	Sophisticated
Percent of Students	35.7	64.3	37.8	62.2	31.6	68.4
Panel B: Cambridge Mechanism						
Percent Assigned to First Choice	78.4	76.2	72.4	69.5	90.2	89.6
Percent Assigned to Second Choice	6.5	12.3	6.6	14.7	6.3	7.6
Percent Assigned to Third Choice	3.3	4.3	4.1	5.9	1.7	1.3
Percent Assigned to Fourth Choice	0.0	1.8	0.0	2.6	0.0	0.3
Percent Assigned to Fifth Choice	0.0	0.5	0.0	0.8	0.0	0.1
Panel C: Deferred Acceptance vs. Cambridge						
Mean Utility DA – Cambridge	−0.024 (0.010)	−0.094 (0.010)	−0.029 (0.014)	−0.13 (0.013)	−0.014 (0.008)	−0.023 (0.007)
Std. Utility DA – Cambridge	0.096	0.125	0.112	0.137	0.047	0.046
Percent DA > Cambridge	22.5 (1.6)	12.8 (1.6)	23.2 (2.0)	9.4 (1.8)	21.0 (2.2)	19.4 (2.3)
Percent DA Cambridge	30.4 (1.2)	30.6 (1.2)	26.6 (1.4)	26.7 (1.4)	38.1 (2.0)	38.2 (2.0)
Percent DA < Cambridge	47.1 (1.6)	56.7 (1.6)	50.2 (1.9)	63.9 (1.9)	40.9 (2.5)	42.4 (2.6)
Percent With Justified Envy	6.4 (0.4)	2.5 (0.2)	8.3 (0.6)	2.7 (0.3)	2.6 (0.4)	2.3 (0.4)

^aPanels A and B present percentages of students assigned to true k th choice. Panel C compares the expected utility difference between Deferred Acceptance and Cambridge Mechanism. Simulations of the Deferred Acceptance mechanism draw other student reports using the estimated utility distribution. We say DA $\approx$ Cambridge if the expected utility is within 10^{-2} miles. Bootstrap standard errors in brackets.

Table XII: Deferred Acceptance versus Cambridge under the mixture model.

7 Short summary

This paper studies how to estimate school preferences when the observed rank lists are strategic. In Cambridge, students face an Immediate Acceptance mechanism, so the report is not just a ranking of tastes. It is a tactical choice over assignment probabilities.

The first empirical step shows that reports respond to incentives. At the geographic boundary where proximity priority changes, the probability of ranking a school first changes discontinuously. The effect is strongest for paid-lunch students and competitive schools. This is direct evidence that rank lists cannot be read as truthful preferences.

The paper then builds a revealed-preference model. Each report is a lottery over school assignments. If a student chooses a report, the expected utility of its lottery must be at least as large as the expected utility of alternative feasible reports. These inequalities restrict the student’s utility vector.

Identification comes from variation in assignment probabilities and from distance as a preference shifter. Estimation proceeds in two steps: first estimate assignment probabilities by simulating the mechanism, then estimate a random utility model using the inequality restrictions implied by optimal reporting.

Empirically, the paper finds that strategic corrections matter. About 83% of students are assigned to their stated first choice, but only about 72% to their true first choice. Under the baseline rational-expectations model, Cambridge’s Immediate Acceptance mechanism slightly beats Deferred Acceptance in average assignment utility, especially for paid-lunch students, though this advantage must be weighed against strategizing costs and fairness concerns.

8 Conclusion

8.1 Core conclusion

Strategic reports are not useless data, but they are not truthful preference lists. They are choices over lotteries. The correct empirical object is the student’s expected-utility comparison across feasible assignment lotteries.

8.2 Economic intuition

Immediate Acceptance rewards students for ranking competitive schools first. A student may avoid a truly preferred school if it is too risky and instead rank a safer school first. This behavior embeds information about preferences, beliefs, and admission probabilities.

8.3 Incremental contribution revisited

The paper provides a bridge between school-choice mechanism design and empirical demand estimation. It shows how to recover preferences in environments where truthful-revelation assumptions fail.

8.4 Implications for school-choice policy

A high rate of assignment to stated first choices is not a sufficient welfare statistic. Policymakers must ask whether students are assigned to their true preferred schools and how much effort, information, and sophistication are required to obtain good outcomes.

8.5 Implications for empirical work

Researchers using rank-order lists from manipulable mechanisms should not estimate standard discrete-choice models as if the rank lists were truthful. Doing so can understate demand for competitive schools and overstate satisfaction with the mechanism.

8.6 Limitations

The welfare analysis does not fully price the cost of strategic thinking or unequal access to information. The identification strategy relies on distance as a valid preference shifter and abstracts from a full model of residential choice.

8.7 Takeaway

The paper’s main lesson is methodological and substantive: when mechanisms are strategic, preferences are hidden behind incentives. But with the right revealed-preference structure, those incentives can be used to recover demand and evaluate market design.